Assignment 5 — PH 4202 'Adv.' Statistical Physics

1. Homogeneous function: A function $f(r)$ is by definition homogeneous if for all values of the parameter λ,

$$f(\lambda r) = g(\lambda) f(r).$$

Show that

$$g(r) = \lambda^p,$$

where p is some constant. Assume $f(r)$ is differentiable.
Show that it further follows that $f(r)$ is itself a power law, that is

$$f(r) = c\, r^a.$$

What are c and a in terms of λ and p?

2. [Problem 16.4. pg 414 Huang] The accompanying graph in the figure below shows the magnetization m vs. magnetic field h for nickel, as measured by Weiss and Forrer 100 years ago in 1926. Some numerical values are also listed in in the figure. [You can also access the data here.] Given that $T_c = 627.2$ K, $\beta = 0.368$, and $\delta = 4.22$, select data from a few temperatures above and below T_c to verify the scaling form via data collapse.

$$m(t,h) \sim |t|^{\beta} \mathcal{F}_{\pm}(h/|t|^{\beta\delta}). \tag{1}$$

3. Equation 1 above is the Widom scaling form for the magnetization we studied in class (e.g., Goldenfeld pg 203). Show that the assumption that $m(t,h)$ is a generalized homogeneous function (GHF) also allows us to write another scaling form

$$m(t,h) = |h|^{\theta} \mathcal{G}_{\pm}(t/h^{\Delta_2}). \tag{2}$$

Detemine θ and Δ_2 in terms of the usual critical exponents β and δ.
A GHF is defined as $f(\lambda^a x, \lambda^b y) = \lambda f(x,y)$.

T	*H*	*M*	*T*	*H*	*M*
678.86	4,160	0.649	614.41	1,820	19.43
	10,070	1.577		3,230	20.00
	17,775	2.776		6,015	20.77
653.31	2,290	0.845		10,070	21.67
	4,160	1.535		14,210	22.36
	6,015	2.215		17,775	22.89
	10,070	3.665	601.14	3,230	25.34
	14,210	5.034		6,015	25.83
	17,775	6.145		10,070	26.40
	21,315	7.211		14,210	26.83
629.53	1,355	6.099		17,775	27.21
	3,230	9.287			
	6,015	11.793			
	10,070	13.945			
	14,210	15.405			
	17,775	16.474			

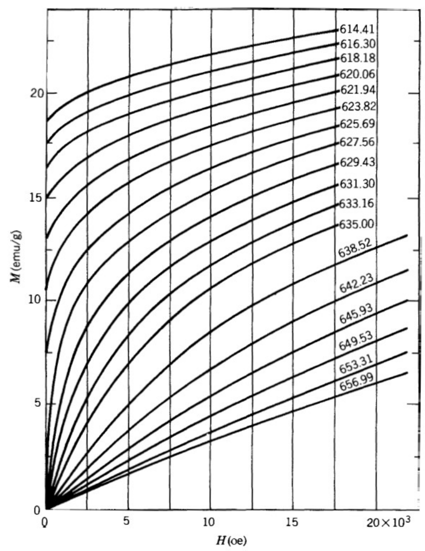

Figure 1: Figure for problem 2. You can also access the data here

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